CPSC-406 Report

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Abstract

Contents

1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$. Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$ (columns: Accepted by $\mathcal{A}_1$? and Accepted by $\mathcal{A}_2$?).

Exercise 1.2

Describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba

3. All the words that contain an odd number of a's and an odd number of b's
4. All the words that contain an even number of a's and an odd number of b's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answers:

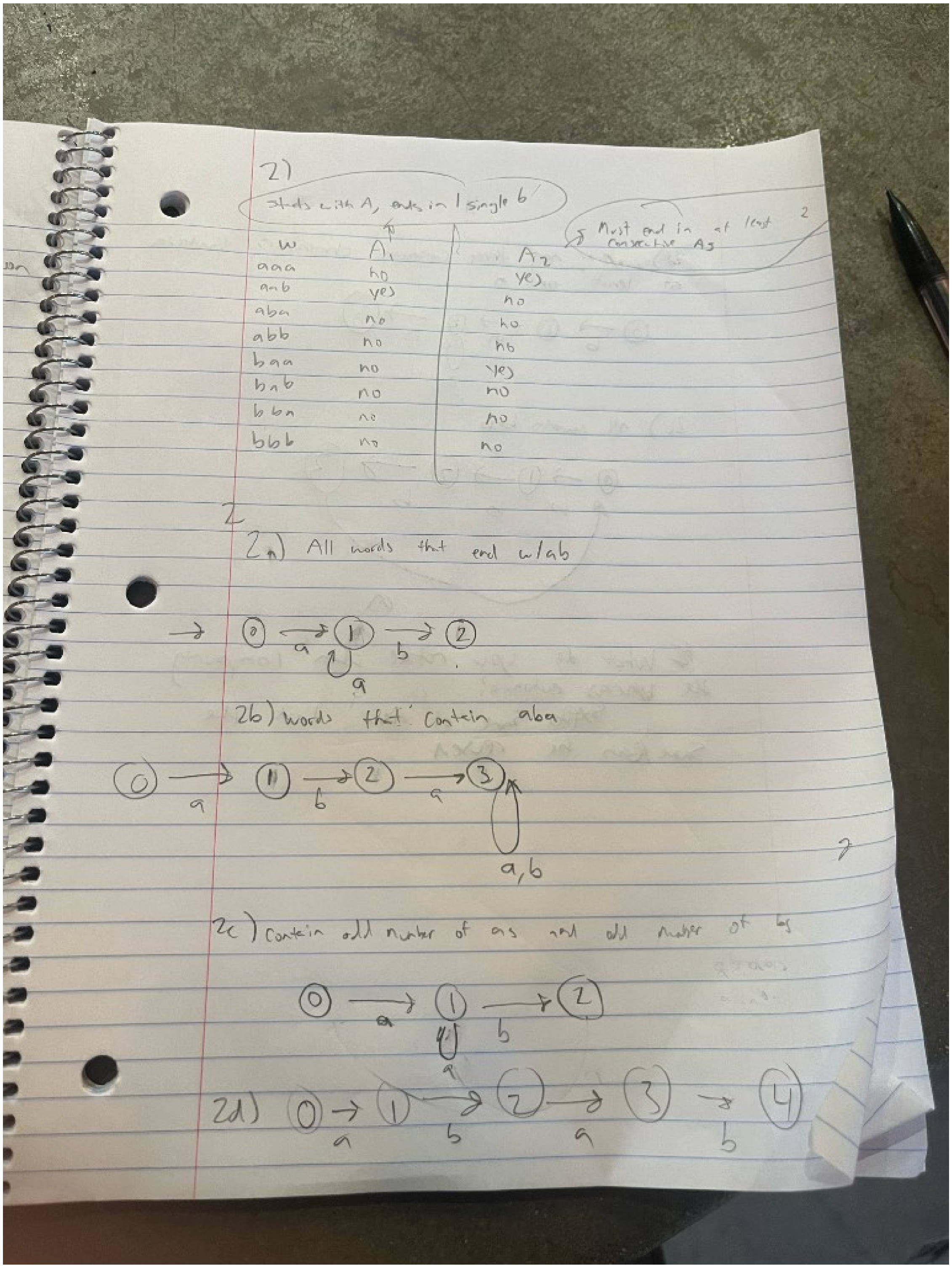

Figure 1: Week 1 answers (table, language conditions, DFA designs 2a–2d).

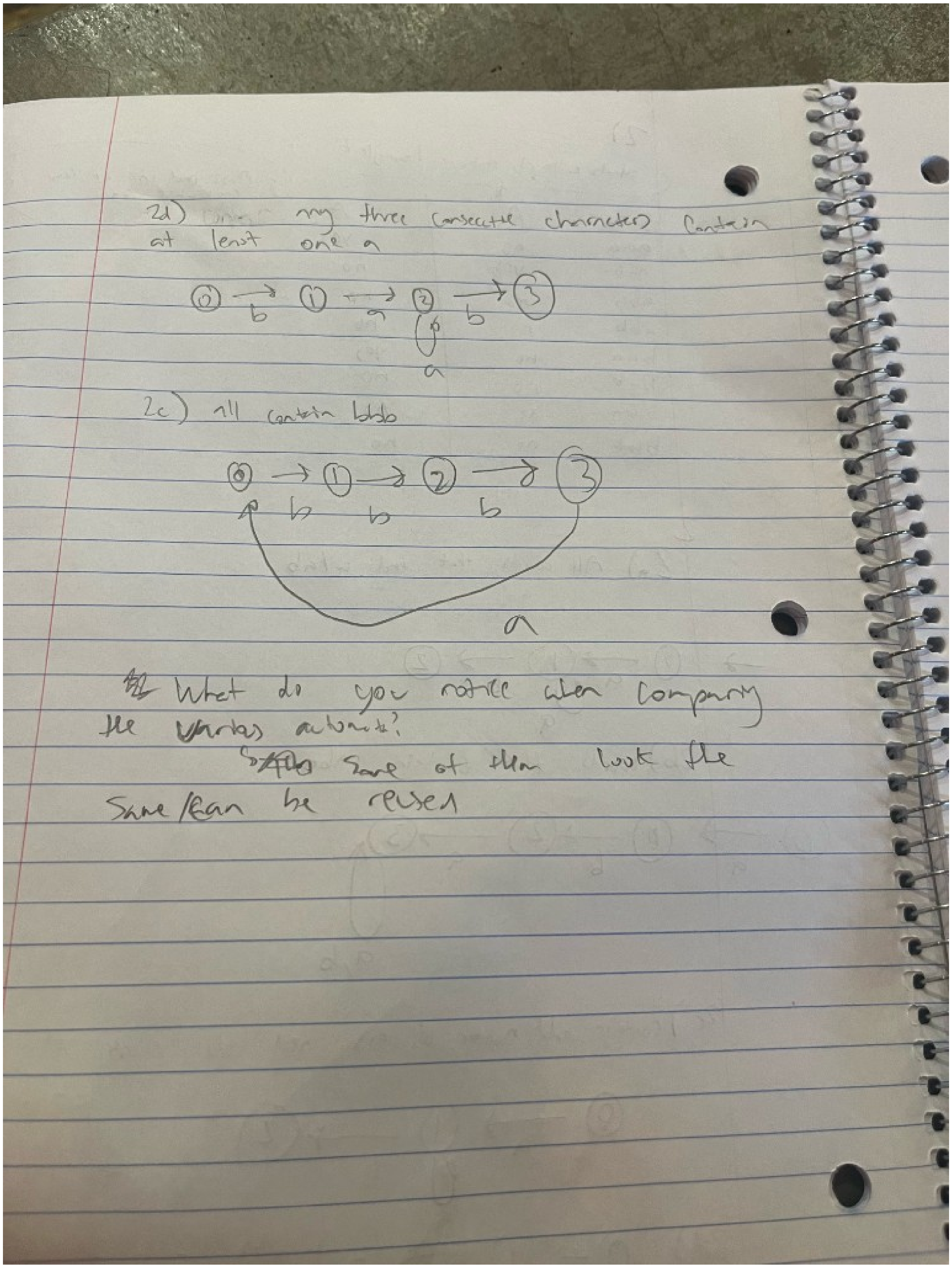

Figure 2: Week 1 answers (DFA designs 2c, 2d; comparison of automata).

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

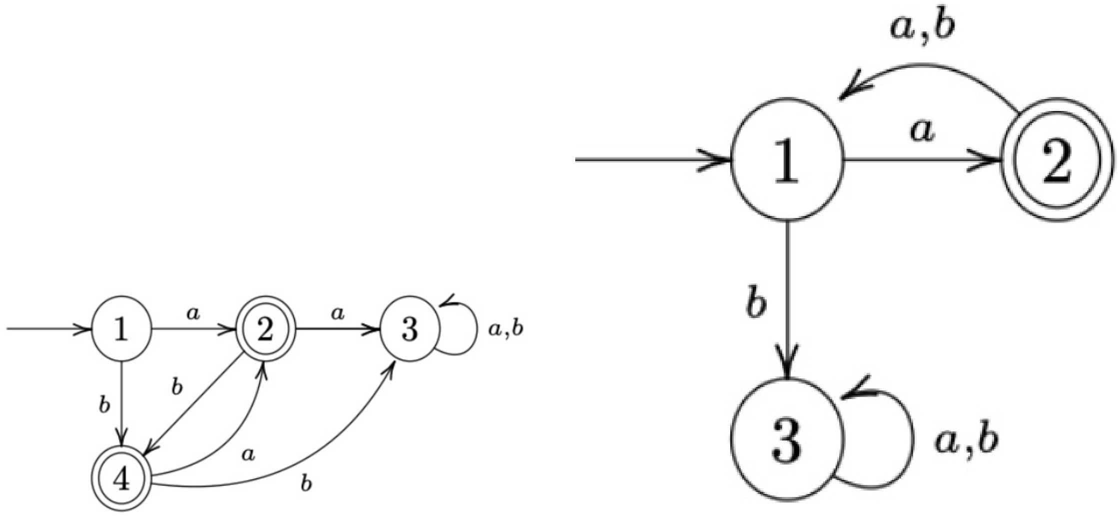

Figure 3: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. **Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.**

   Answer:

   - $L(\mathcal{A}^{(1)})$: Words include a, b, ba, bab, $baba$, ab, aba, $abab$, etc. I can see that each character must alternate based on its successive character.
   - $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a, and must add to its word length in increments of 2, every other character being a.

2. **Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).**

   Answer: Below is the intersection automaton diagram.

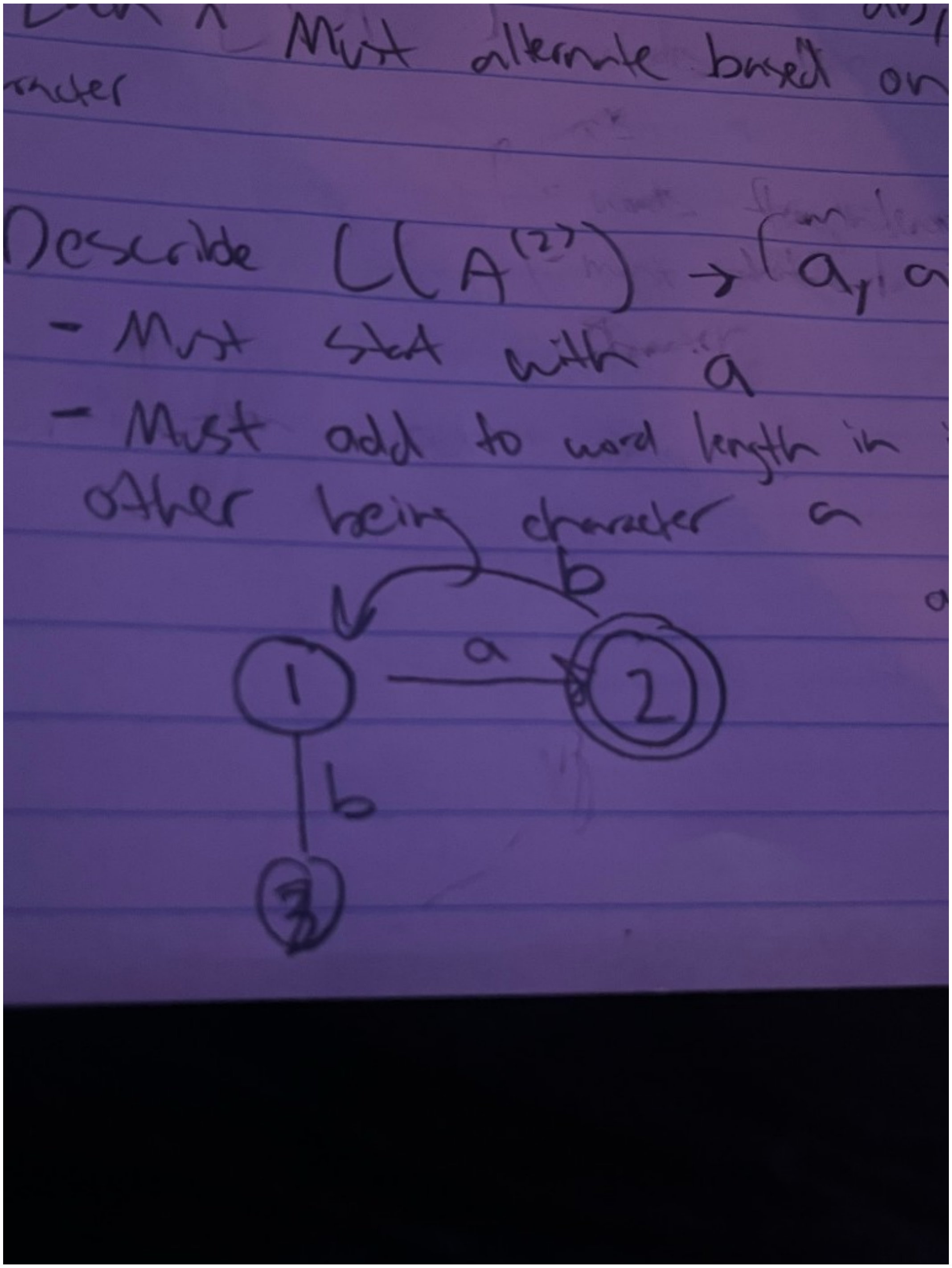

Figure 4: Intersection automaton $\mathcal{A}$ (hand-drawn diagram).

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$**.**

   Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a, b, ba, bab, $baba$ and ab, aba, $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a—so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$, leaving us with a, ab, aba, etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: a, aba, $ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

   Answer: We can allow for both word combinations to be accepting and add more state transitions.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

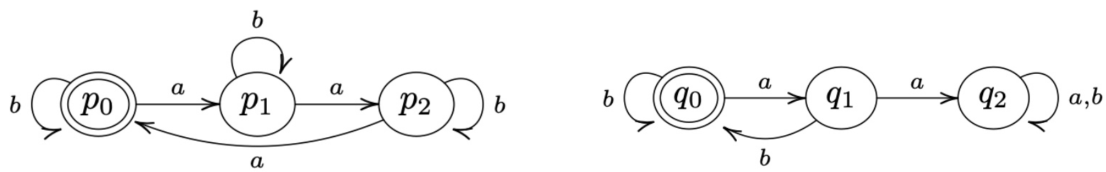

Figure 5: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\overline{L} := \Sigma^* \setminus L$.)

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

   Answer:

Figure 6: $L(\mathcal{B}^{(1)})$ rules: include 3 a's at a time, any amount of b's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a, any amount of b's. DFA diagram for language 2.

2. **Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).**

   Answer: Below is the intersection automaton diagram.

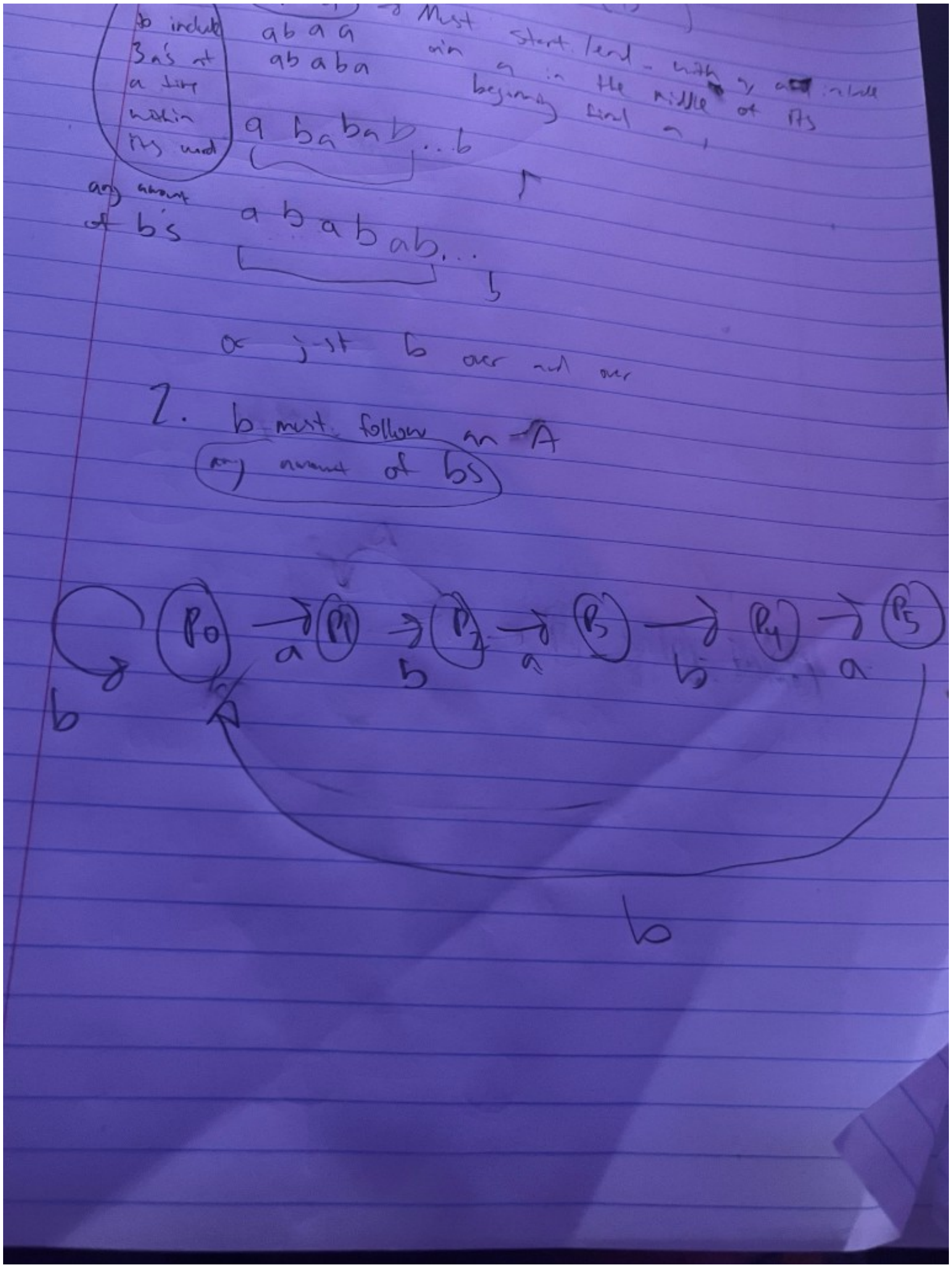

Figure 7: $L(\mathcal{B}^{(1)})$ rules: include 3 a's at a time, any amount of b's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a, any amount of b's. DFA diagram for language 2.

3. **Argue that** $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$**.**

   Answer: Both automata have the rule that any amount of b's is allowed. But we must combine the rules regarding a's. $L(\mathcal{B}^{(1)})$ requires 3 a's to be included at a time per word, while $\mathcal{B}^{(2)}$ requires the rule that a must be followed by a b. This gives us the conclusion that words that do include an a must come in pairs of 3 a's with b following the last a.

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

   Answer: We must flip the accepting states of automaton 1 to include: no amount of b's AND words that include 3 a's at a time only.

Exercise 3

Let $\mathcal{A}$ be a DFA and q a state of $\mathcal{A}$ such that from q, every input symbol a leaves us in q (i.e. $\delta(q, a) = q$ for all a). Show by induction on the length of the input that for any input string w, we still end up in q (i.e. $\delta^*(q, w) = q$).

Answer: We show it by induction on the length of w. (Here $\delta^*(q, w)$ is just the state we end up in after starting at q and reading the whole string w.)

When the input has length 0, the string is empty. So we don't read anything and we stay at q. So $\delta^*(q, w) = q$ in that case.

Now suppose we've already shown that for every string of length n, we end up back at q. Take a string w of length $n + 1$. We can write it as $w = va$ (some string v of length n plus one symbol a). After reading v we're in $\delta^*(q, v)$, and by our assumption that's q. Then we read a, and we're in $\delta(q, a)$, which is q by the given rule. So after reading w we're still in q, i.e. $\delta^*(q, w) = q$.

So by induction, for all input strings w we have $\delta^*(q, w) = q$.

Question

If you build the product automaton for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ but change the accepting set to get the *union* $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$ instead of the intersection, how many states must be accepting?

2.3 Week 3

Homework 1 (NFA)

1. Describe the language accepted by $\mathcal{A}$ using a regular expression.
2. Specify the NFA $\mathcal{A}$ in the form $(Q, \Sigma, \delta, q_0, F)$.
3. Find all paths in $\mathcal{A}$ for the word $w = abab$; represent in a diagram.

Answer (NFA specification). Here is how I write the NFA $\mathcal{A}$ formally:

$$Q = \{0, 1, 2, 3\}, \qquad \Sigma = \{a, b, c\}, \qquad q_0 = 0, \qquad F = \{2, 3\}.$$

For the transition function $\delta : Q \times \Sigma \to \mathcal{P}(Q)$ I use

$$\begin{aligned} &\delta(0, a) = \{0, 1\}, \quad &&\delta(0, b) = \{0\}, \quad &&\delta(0, c) = \emptyset, \\ &\delta(1, b) = \{2\}, &&\delta(1, c) = \{2\}, \end{aligned}$$

and any transition not listed goes to the empty set (i.e. there is no move).

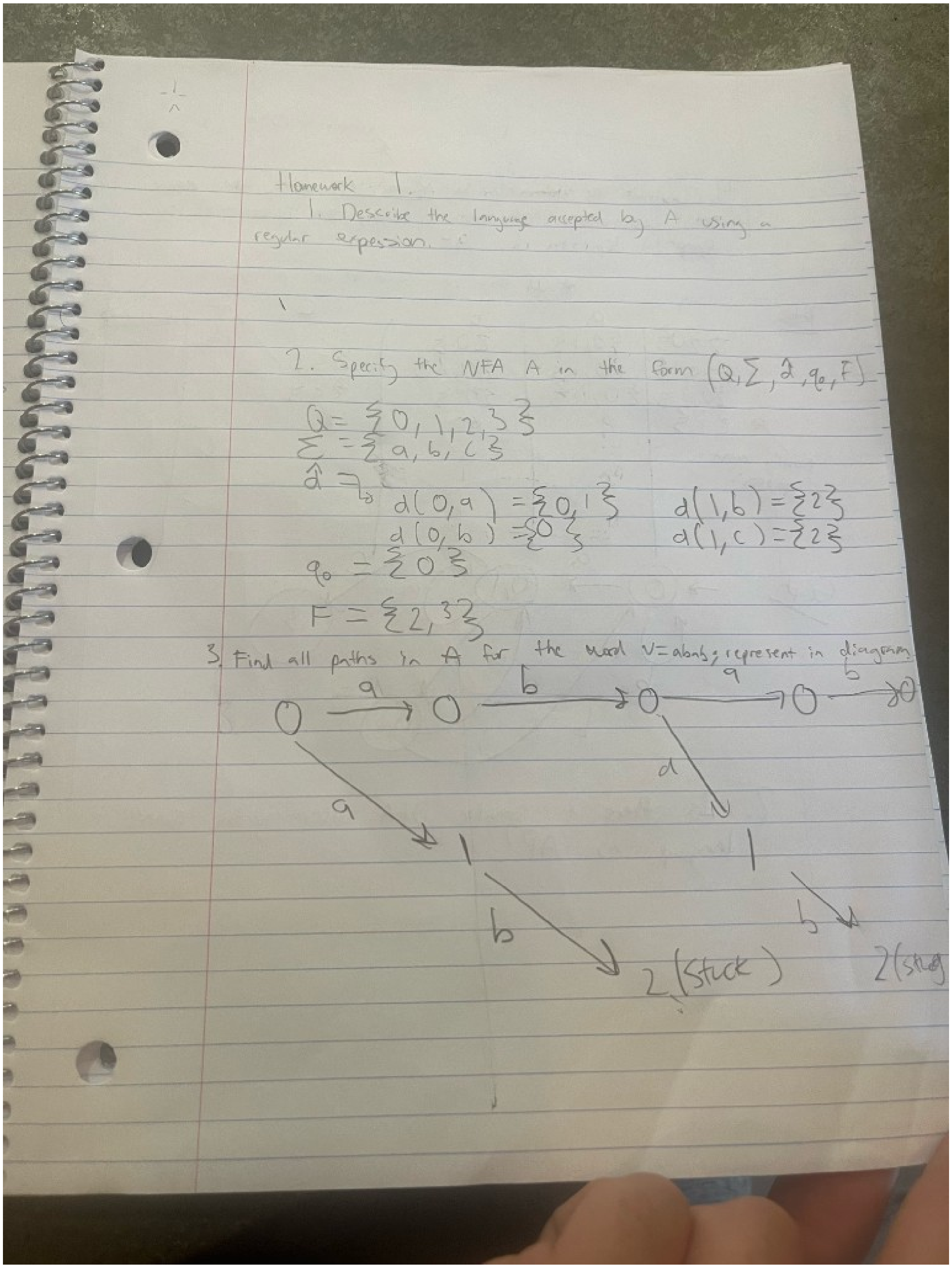

Figure 8: Homework 1: NFA specification ($Q = \{0, 1, 2, 3\}$, $\Sigma = \{a, b, c\}$, $q_0 = 0$, $F = \{2, 3\}$) and all paths for $w = abab$.

Homework 2 (Determinization)

Construct the determinization $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction. Create both a transition table and a graphical depiction of $\mathcal{A}^D$.

Transition table for $\mathcal{A}^D$ over $\{a, b, c\}$. Starting from the NFA above, the reachable subset-states under the power set construction are

$$\{0\},\ \{0,1\},\ \{0,1,2\},\ \{0,2,3\},\ \{2,3\},\ \emptyset.$$

I call a subset accepting when it contains either 2 or 3.

State S	on input a	on input b	on input c
$\{0\}$	$\{0,1\}$	$\{0\}$	$\emptyset$
$\{0,1\}$	$\{0,1\}$	$\{0,2\}$	$\{2\}$
$\{0,1,2\}$	$\{0,1\}$	$\{0,2\}$	$\{2\}$
$\{0,2\}$	$\{0,1\}$	$\{0,2\}$	$\emptyset$
$\{2\}$	$\emptyset$	$\{2\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

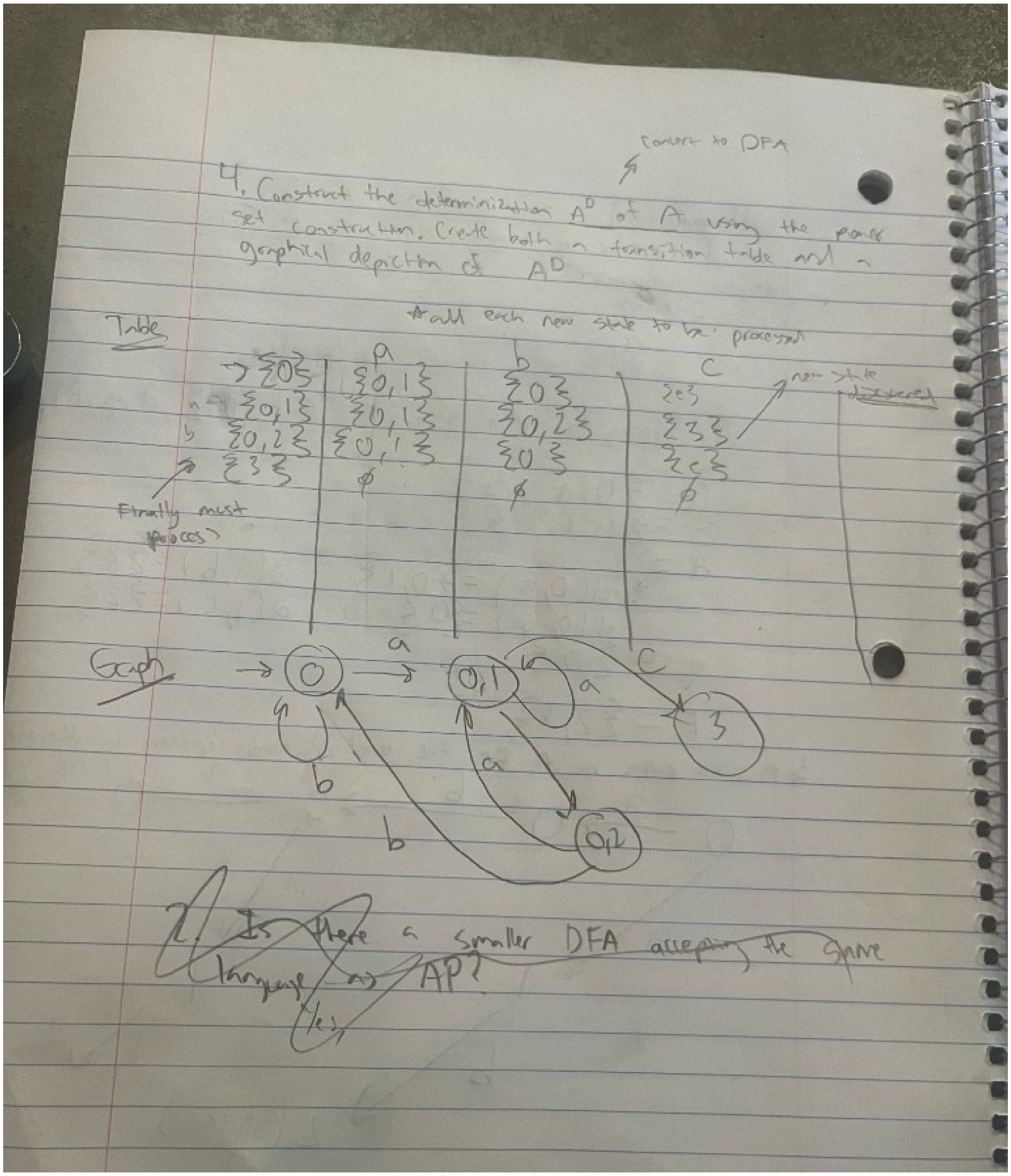

Figure 9: Graph of $\mathcal{A}^D$ (photo); the transition table is typeset separately in LaTeX.

Homework 3 (DFA and minimization)

Determine $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction, now with alphabet $\{0,1\}$. Create both a transition table and a graphical depiction of $\mathcal{A}^D$. Then decide whether there can be a smaller DFA accepting the same language.

Transition table over $\{0,1\}$. From the NFA given in the homework, the reachable subset-states are

$$\{q_0\},\ \{q_0,q_1\},\ \{q_0,q_1,q_2\},\ \{q_0,q_1,q_2,q_3\},\ \{q_0,q_3\}.$$

The determinized DFA $\mathcal{A}^D$ has the following transitions:

State S	on input 0	on input 1
$\{q_0\}$	$\{q_0,q_1\}$	$\{q_0\}$
$\{q_0,q_1\}$	$\{q_0,q_1,q_2\}$	$\{q_0,q_1\}$
$\{q_0,q_1,q_2\}$	$\{q_0,q_1,q_2,q_3\}$	$\{q_0,q_1\}$
$\{q_0,q_1,q_2,q_3\}$	$\{q_0,q_1,q_2,q_3\}$	$\{q_0,q_1\}$
$\{q_0,q_3\}$	$\{q_0,q_1,q_2,q_3\}$	$\{q_0,q_1\}$

Any subset that contains the original accepting state(s) is accepting in $\mathcal{A}^D$.

Smaller DFA? Yes. In this table the last two rows (for $\{q_0,q_1,q_2,q_3\}$ and for $\{q_0,q_3\}$) have exactly the same outgoing transitions and the same accepting/non-accepting status. That means these two states are equivalent and can be merged into a single state, giving a smaller DFA that still recognizes the same language.

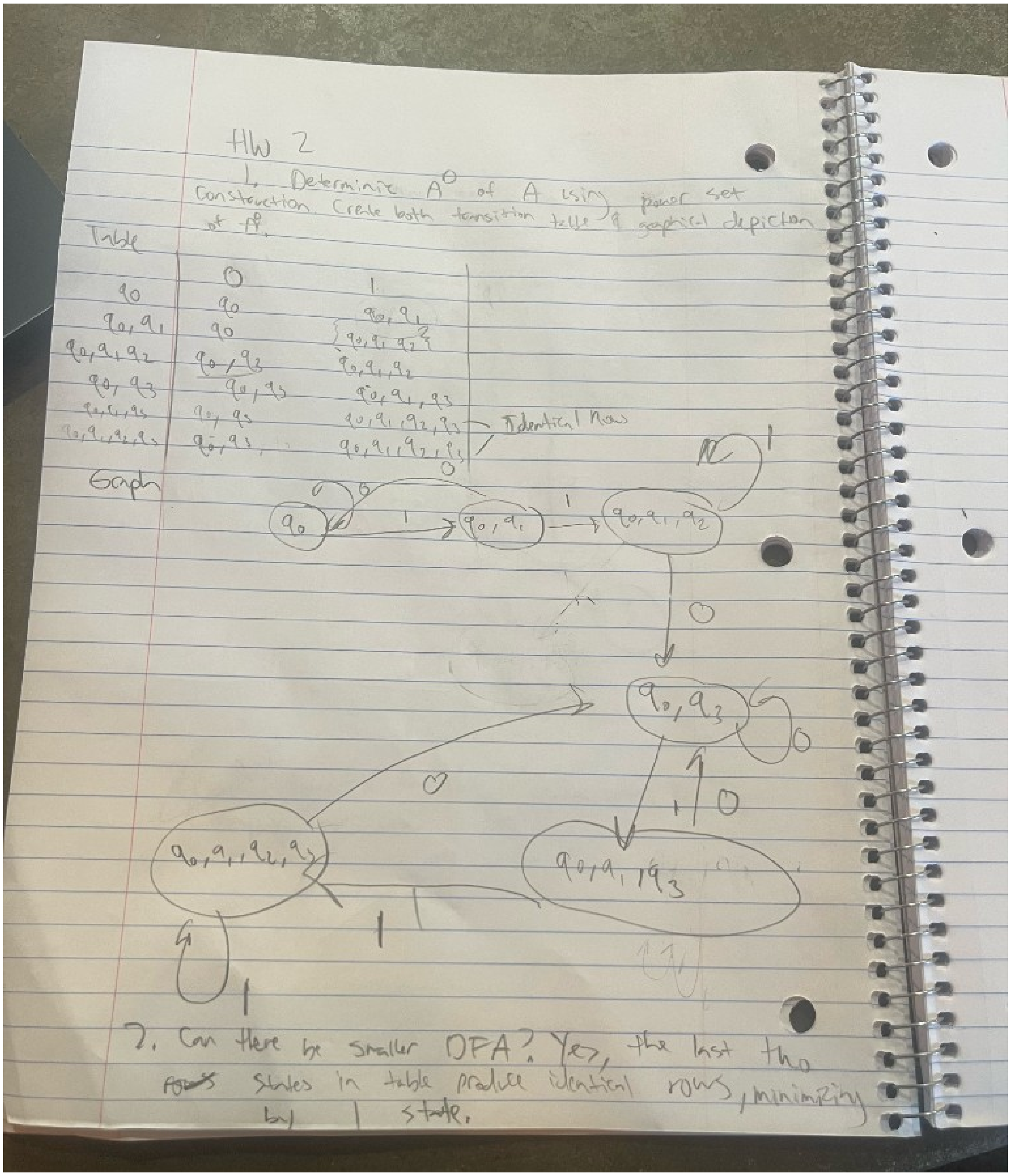

Figure 10: State diagram of $\mathcal{A}^D$ for Homework 3 (photo of the graph only).

Question

After building $\mathcal{A}^D$ from an NFA $\mathcal{A}$ using the power set construction, when can we get a smaller DFA that still accepts the same language?

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.